

Supplementary materials

Table S1 The number of neurons recorded in each session for WT and 5xFAD groups.

5xFAD				
Mouse index	session 1	session 2	session 3	session 4
1	411	336	—	408
16	295	276	—	288
117	36	73	78	94
118	27	17	28	—
123	80	68	128	—
WT				
3	378	381	295	564
5	413	—	574	352
7	—	196	627	420
102	173	72	—	47
106	106	341	82	—
109	144	97	125	108
110	284	244	—	275

Cross-validation

During fine-tuning, cross-validation was performed to test the hypothesis regarding the influence of individual session quality on the final results.

Loss function curves during network training for each fold are shown in [Figure S1](#). The F1 score curve is shown in [Figure S2](#). The plots are presented in a table, where each cell contains the curve for a specific fold; columns are organized by the transgenic subject ID, and rows by the wild-type subject. These plots demonstrate that the network trains stably, with the exception of some folds involving subject 106. In several cases, an increase in the loss function is observed during the initial epochs, accompanied by a corresponding drop in the F1 score. The average accuracy and standard deviation for each session of the validation sample calculated for the entire cross-validation experiment is shown in the [Table S2](#).

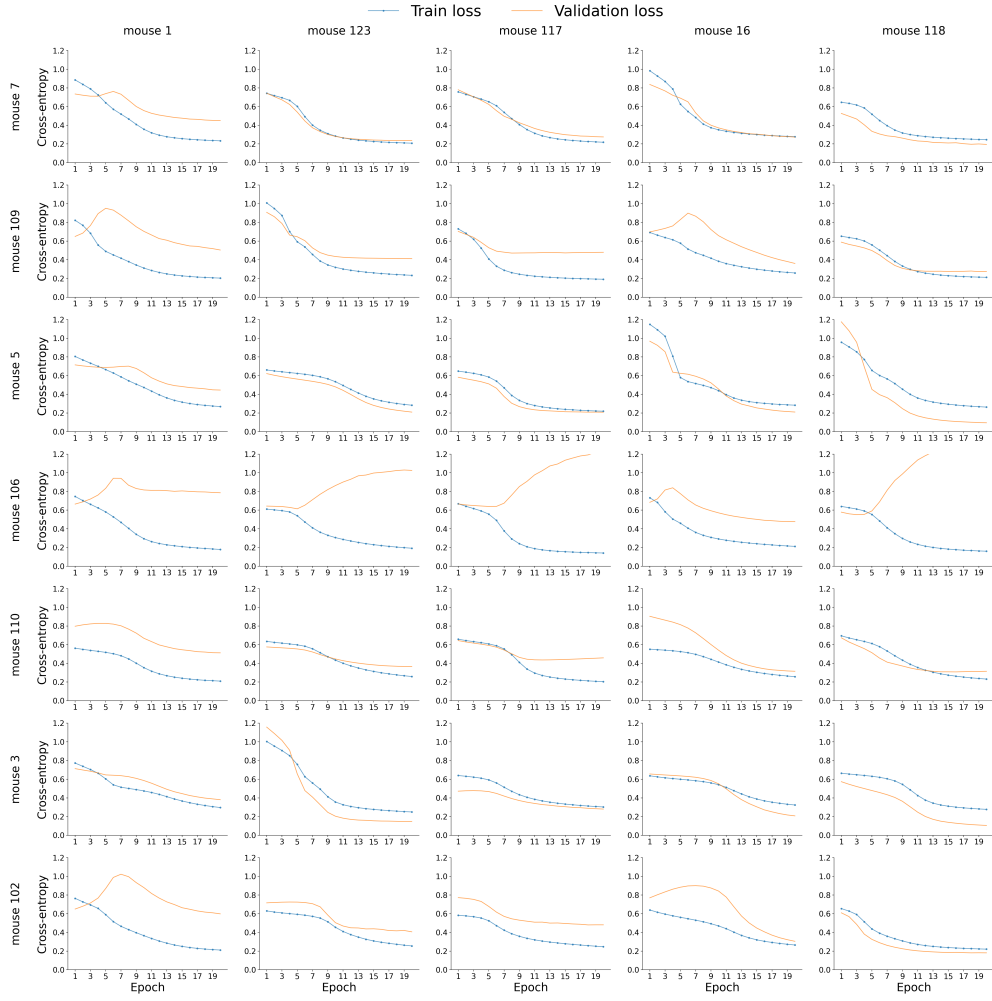


Fig. S1 Curves of the loss function for each fold in the cross-validation experiment. Each graph shows the loss functions by epochs for the training and validation subsample. Each row is signed on the left and corresponds to one wild-type individual. Each column is signed at the top and corresponds to a transgenic individual. The column signature and the lines of each graph show which two individuals form the validation sample.

In this experiment, 0.5 was chosen as the threshold value for the following reasons. Each activity matrix actually represents the activity of one neuron, provided with some additional statistics about its environment. Thus, the proportion of matrices obtained from one session to a specific class can act as an indirect metric of the model’s confidence in predicting the entire session. If more than half of the data from one session is assigned by the model to a specific class, then we can say that the model as a whole assigned the session to this class. It can be seen from them that, for example, session 2 of individual 106 consistently shows a result below 0.5 in terms of prediction accuracy, while sessions 1 and 3 consistently give a result above 0.5. For the remaining sessions, there is also a fairly stable behavior. Either the session is classified correctly or incorrectly on all the presented folds. Moreover, for a large number of sessions, the same pattern of the training schedule is seen on all folds where it was included in the validation sample. Thus, the hypothesis of a large influence of the initial data on the prediction of the model is confirmed. Regardless of the data breakdown into samples, the model makes similar predictions for the same sessions. the epoch for each session of the validation sample. Such data is especially useful.

Table S2 Mean and standard deviation for each session and mouse, ordered by mouse index.

5xFAD				
Mouse index	session 1	session 2	session 3	session 4
1	0.9913 $\pm$ 0.0048	0.1739 $\pm$ 0.0056	—	0.9252 $\pm$ 0.0527
16	0.8478 $\pm$ 0.0636	0.9569 $\pm$ 0.0311	—	0.9986 $\pm$ 0.0021
117	0.0773 $\pm$ 0.0318	0.1724 $\pm$ 0.0320	0.9688 $\pm$ 0.0309	0.8917 $\pm$ 0.0593
118	0.4925 $\pm$ 0.0115	0.3259 $\pm$ 0.1186	0.9474 $\pm$ 0.0182	—
123	0.6136 $\pm$ 0.1004	0.1741 $\pm$ 0.0649	0.9959 $\pm$ 0.0027	—
WT				
3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9977 $\pm$ 0.0018	0.9980 $\pm$ 0.0007
5	1.0000 $\pm$ 0.0000	—	0.9994 $\pm$ 0.0002	1.0000 $\pm$ 0.0000
7	—	0.9709 $\pm$ 0.0301	0.9878 $\pm$ 0.0038	0.9805 $\pm$ 0.0069
102	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	—	1.0000 $\pm$ 0.0000
106	0.9615 $\pm$ 0.0185	0.2387 $\pm$ 0.0635	0.7131 $\pm$ 0.0485	—
109	0.9302 $\pm$ 0.0492	0.9732 $\pm$ 0.0097	0.9921 $\pm$ 0.0064	0.8916 $\pm$ 0.0512
110	0.9997 $\pm$ 0.0003	0.9999 $\pm$ 0.0002	—	0.5965 $\pm$ 0.0729

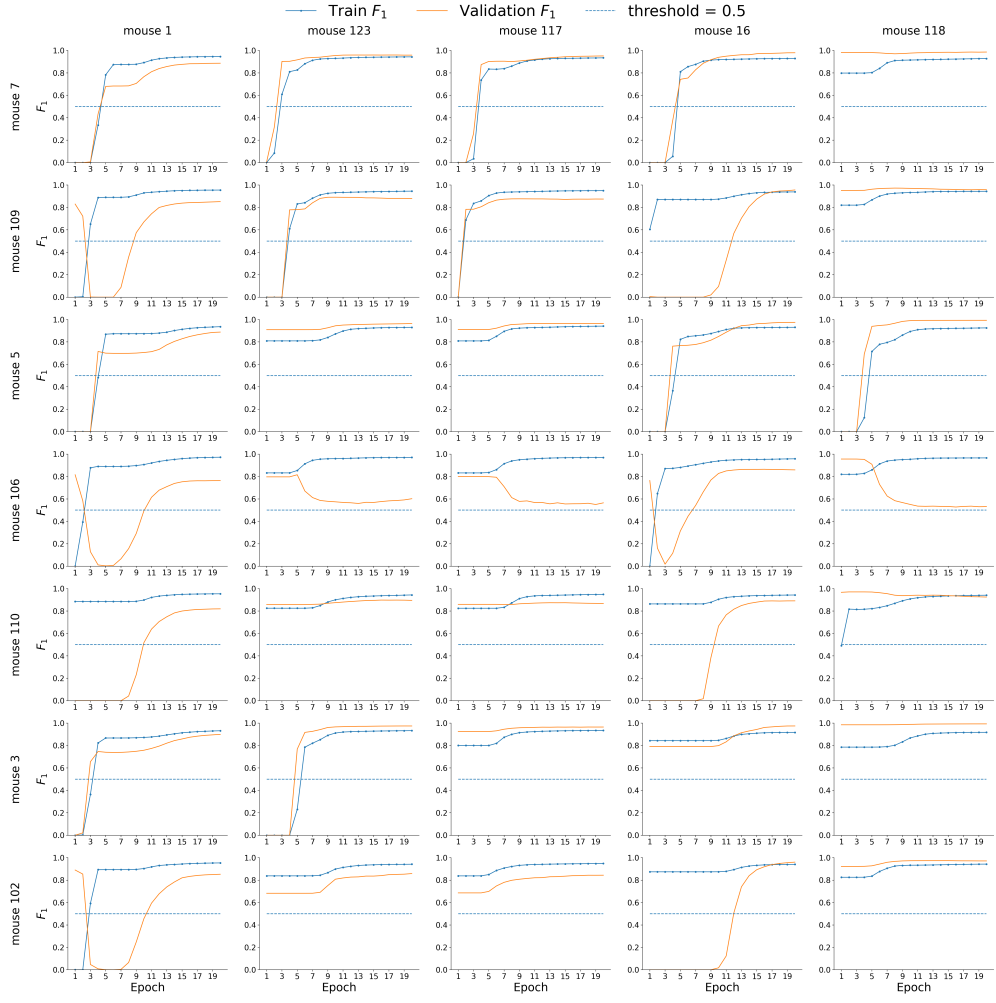


Fig. S2 Curves of the F1 score for each fold in the cross-validation experiment. Each graph shows the F1 score by epochs for the training and validation subsample. Each row is signed on the left and corresponds to one wild-type individual. Each column is signed at the top and corresponds to a transgenic individual. The column signature and the lines of each graph show which two individuals form the validation sample.

For a more detailed analysis, data was obtained on the accuracy of prediction for each in situations where there is first a drop in quality across the entire sample, and then an increase. Graphs of prediction accuracy are shown in [Figure S3](#).

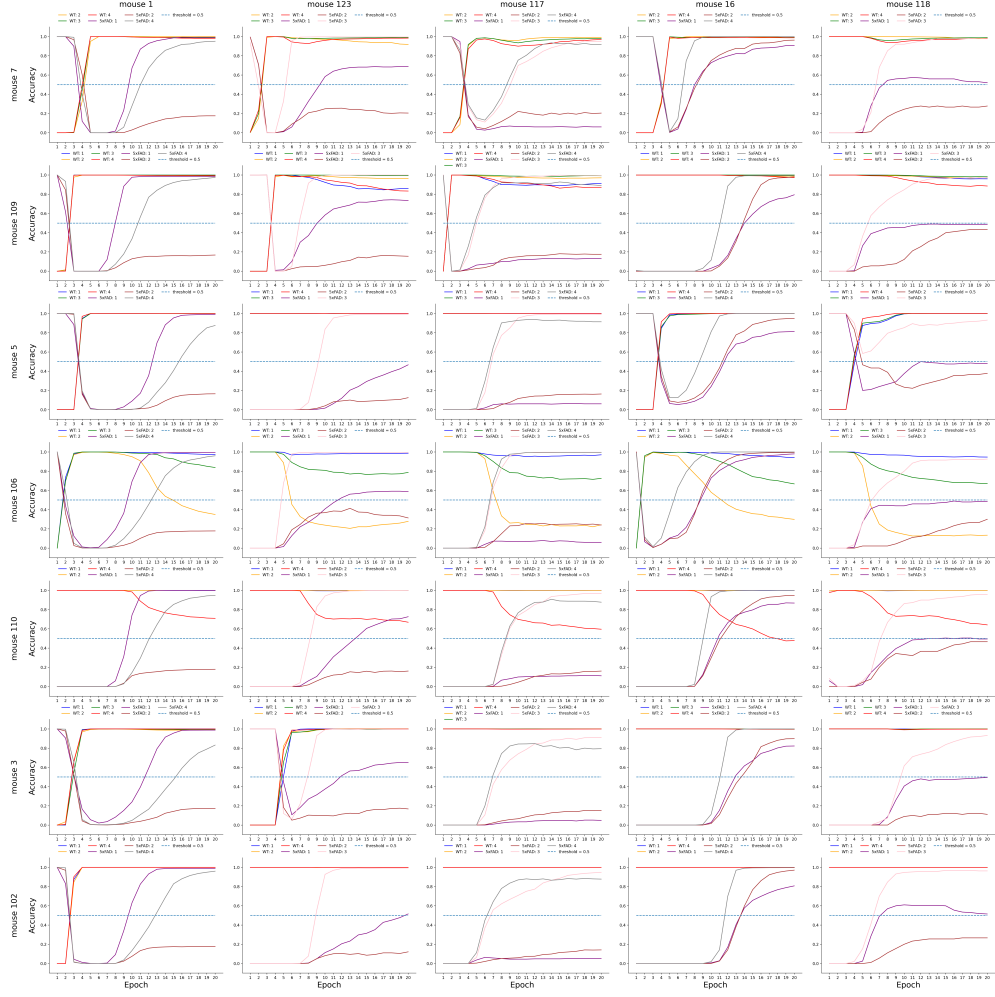


Fig. S3 Classification accuracy across validation sessions for each fold. On the left, the wild-type subject is indicated, and on the top, the subject with the Alzheimer's disease model whose data constitute the validation set. Each graph shows the dependence of classification accuracy on the epoch number for each session participating in the fold. The colors for each session are marked in the legend for each chart. The colors of the charts are the same for the respective sessions for each mouse. The threshold is equal to 0.5.

Interpretation methods

In Figure S4, the confusion matrices obtained by zeroing out individual components of the activity matrix are shown.

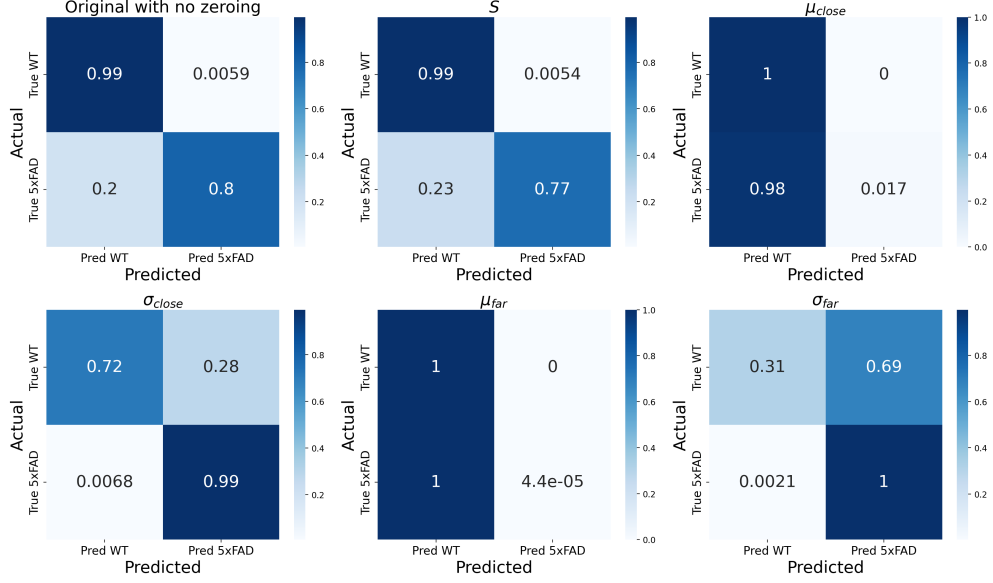


Fig. S4 Confusion matrices for the model outputs when individual components of the activity matrix are zeroed out. The color gradient on the color bar indicates results normalized to the true value.

To estimate the importance of temporal segments, we employed an adapted attention rollout method, while also testing simplified and extended variants. Two modifications were considered. The first involved using entropy within a single attention layer to reduce the dominance of diagonal attention matrices:

$$W_l = softmax(\frac{1}{L} \sum_{i=1}^L (- \sum_{j=1}^L A_{lij}^h \log_2 (A_{lij}^h + 10^{-12})))$$

$$\overline{A}_l^h = W_l^h \cdot A_l^h$$

Instead of the standard averaging over attention heads within a layer, a weighted average was applied.

The second modification involved using gradients as weights, not for entire matrices, but for individual elements. A similar approach was used in (Chefer et al., 2021), where relevance weights were applied instead of attention matrices. The gradient-based weighting is expressed as:

$$\overline{A}_l^h = \nabla A_l^h \odot A_l^h$$

We compared variants using entropy alone and entropy combined with gradients. The most notable behavior, however, was observed when using only the pooling vector weights to estimate temporal importance, as shown in [Figure S5](#), where methods are compared in the top-k zeroing experiments described in Results.

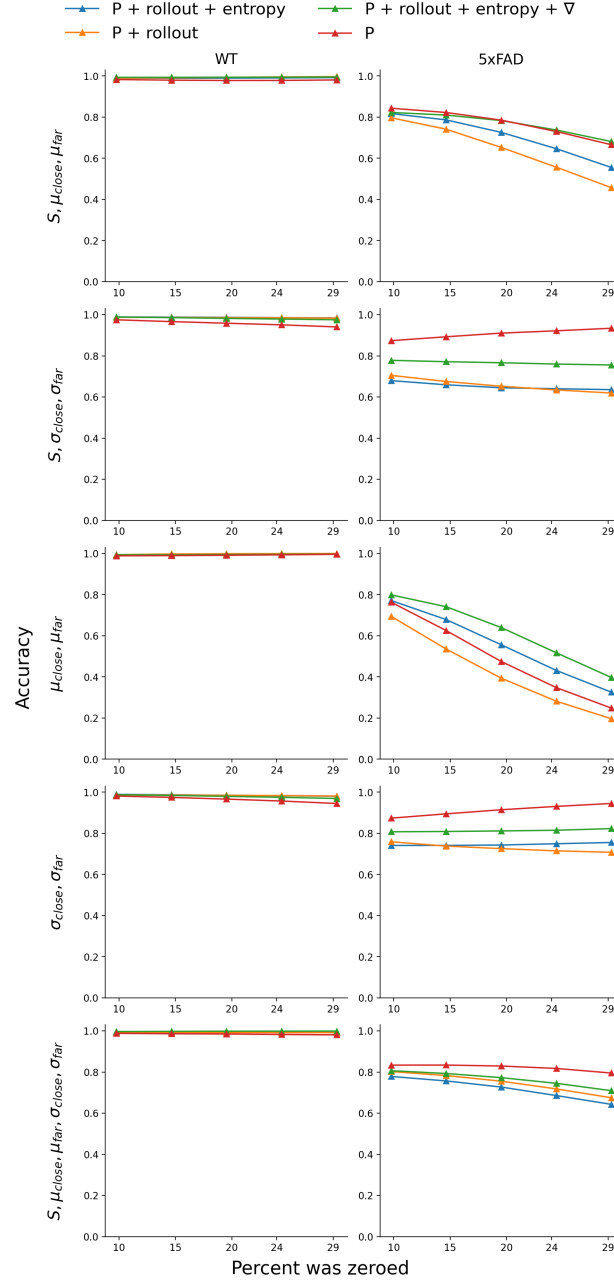


Fig. S5 Comparison of the model's prediction accuracy for both classes when zeroing out the top-k temporal segments of activity matrix component groups based on their importance using different interpretation methods: P + rollout + entropy – a combination of pooling weights and entropy-weighted attention rollout; P + rollout – a combination of pooling weights and unweighted attention rollout; P + rollout + entropy + ∇ – a combination of pooling weights, entropy-weighted attention rollout, and gradients; P – pooling weights alone.